

Muhammad Mubashir

+92-333-7588703 | mmubashirmemon@gmail.com
linkedin.com/in/mmubashir72 | github.com/mubashir72

SUMMARY

Software Engineering graduate and MS Artificial Intelligence student at FAST NUCES with hands-on experience developing Python backend services and AI-powered applications. Built Generative AI, Large Language Model (LLM), and Retrieval-Augmented Generation (RAG) workflows using prompt engineering, LangChain, and AI APIs. Applied object-oriented programming, RESTful APIs, SQL databases, clean architecture, Git, automated testing, and continuous integration across development work and projects.

EDUCATION

FAST NUCES <i>Master of Science in Artificial Intelligence, CGPA: 3.44/4.00</i>	Islamabad, PK <i>Aug. 2025 – Present</i>
MUET <i>Bachelor of Engineering in Software Engineering</i>	Jamshoro, PK <i>Nov. 2020 – Nov. 2024</i>

TECHNICAL SKILLS

Programming: Python, Java, C++, JavaScript, SQL
Generative AI: LLMs, RAG, Prompt Engineering, LangChain, Groq API, OpenAI API
AI/ML Libraries: PyTorch, TensorFlow, scikit-learn, NumPy, pandas, Matplotlib
Backend & Databases: FastAPI, PostgreSQL, SQLite, MongoDB, Firebase Authentication, Cloud Firestore
Frontend & Interfaces: React, HTML, CSS, Tailwind CSS, Streamlit, Gradio
Software Engineering: Object-Oriented Programming (OOP), Data Structures & Algorithms, Software Design, RESTful APIs, Clean Architecture, Repository Pattern
Development & Quality: Git, GitHub, Docker, Docker Compose, pytest, GitHub Actions
Professional Skills: Problem-Solving, Teamwork, Communication, Time Management, Attention to Detail

EXPERIENCE

Backend AI Engineering Intern <i>FlyRank AI</i>	Aug. 2026 – Present <i>Remote</i>
- Developed RESTful APIs and asynchronous backend services using Python and FastAPI, following modular software engineering practices. - Applied clean architecture and the repository pattern to separate API routes, business logic, and database access layers. - Integrated SQLite and PostgreSQL databases using SQL to implement persistent application data storage. - Configured Docker and Docker Compose environments for backend services and PostgreSQL with persistent volumes. - Used Git and GitHub for version control and iterative software development.	
AI Engineering Intern <i>Brinicle AI</i>	July 2026 <i>Remote</i>
- Worked on AI engineering tasks involving LLMs, Generative AI, and practical AI application development. - Conducted technical experiments and documented findings from AI/ML development work.	
WordPress Developer Intern <i>KGT Global</i>	July 2025 – Aug. 2025 <i>Karachi</i>
- Developed and customized responsive WordPress websites using Divi Builder, Elementor, PHP, JavaScript, HTML, and CSS. - Implemented features, resolved bugs, and optimized website performance and user experience.	

PROJECTS

PaperSensei | *Python, Streamlit, Groq, Firebase* | [Live Demo](#) | [GitHub](#)

- Built an AI-powered study assistant for a hackathon that converts PDF textbooks and notes into conceptual multiple-choice quizzes and provides context-aware tutoring using LLMs and prompt engineering.
- Implemented adaptive quiz difficulty based on topic-level performance, with simpler explanations and follow-up questions after incorrect answers.
- Integrated Firebase Authentication and Cloud Firestore for email verification, saved study sessions, and conversation persistence, with user-specific database rules.
- Processed PDFs using pdfplumber to preserve page references and support section selection. Added past-paper topic analysis and downloadable CSV progress reports.
- Implemented automated tests with pytest and continuous integration through GitHub Actions.

YouTube Video Q&A System | *Python, LangChain, OpenAI, Pinecone, Whisper*

- Developed an end-to-end Retrieval-Augmented Generation (RAG) application for natural-language question answering over YouTube videos.
- Built transcription, text chunking, embedding, and semantic retrieval pipelines using OpenAI Embeddings, Pinecone, and LangChain.
- Integrated a retriever and LLM workflow with a Gradio interface for video input and context-aware answers.
- Implemented caching and vector-store optimizations to reduce redundant processing and improve retrieval speed.

Medical Chest X-ray Analysis Agent | *Python, PyTorch, LLMs*

- Extended an existing medical AI agent framework for chest X-ray analysis as part of a research project.
- Integrated Grad-CAM explainability into the analysis pipeline to improve model interpretability.
- Evaluated and compared LLMs for diagnostic report generation, and extended the codebase to support experiments and benchmarking.

CERTIFICATIONS

Machine Learning Specialization | DeepLearning.AI, Coursera

PyTorch: Fundamentals | DeepLearning.AI, Coursera

Fundamentals of Building AI Agents | IBM, Coursera